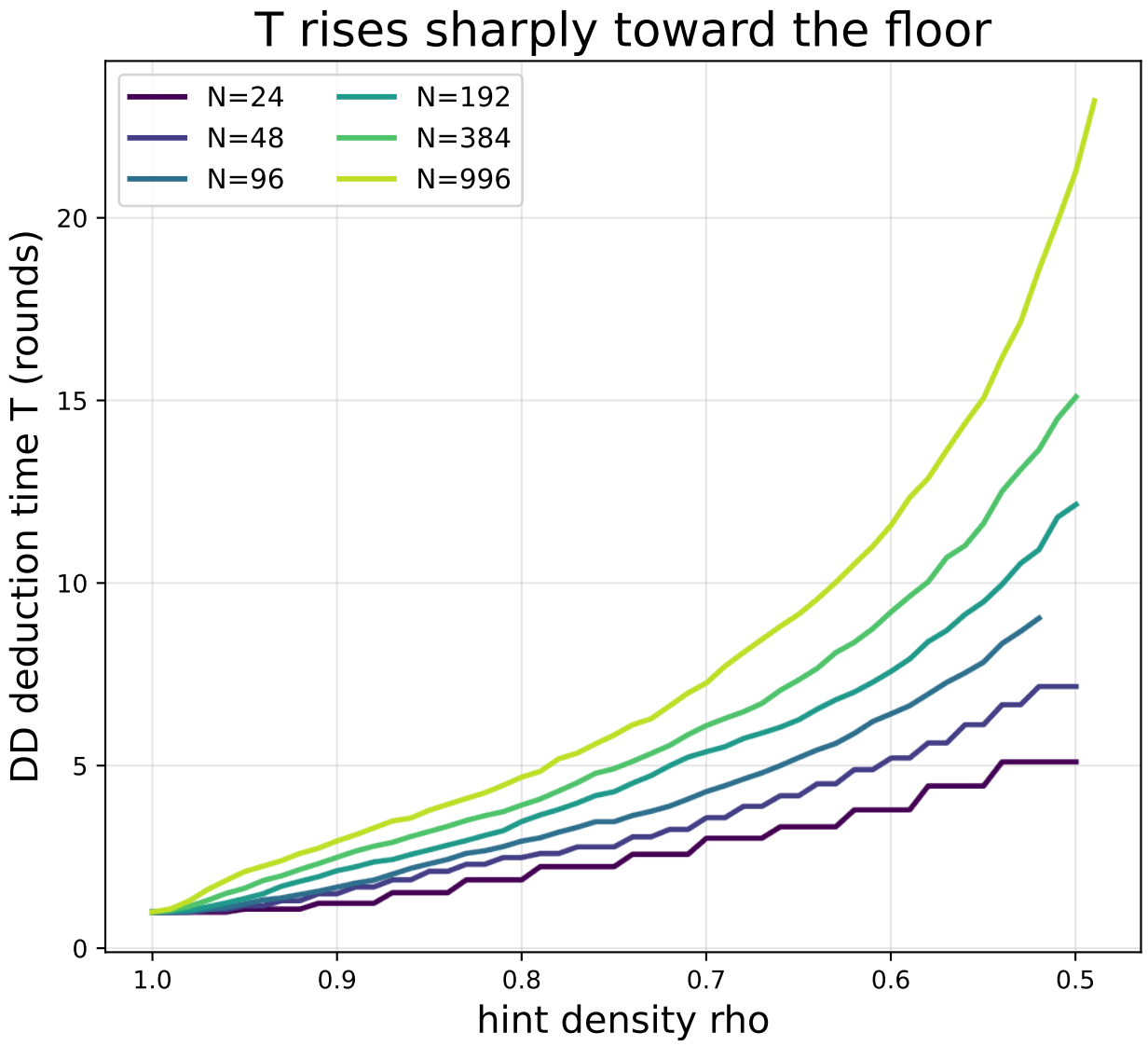
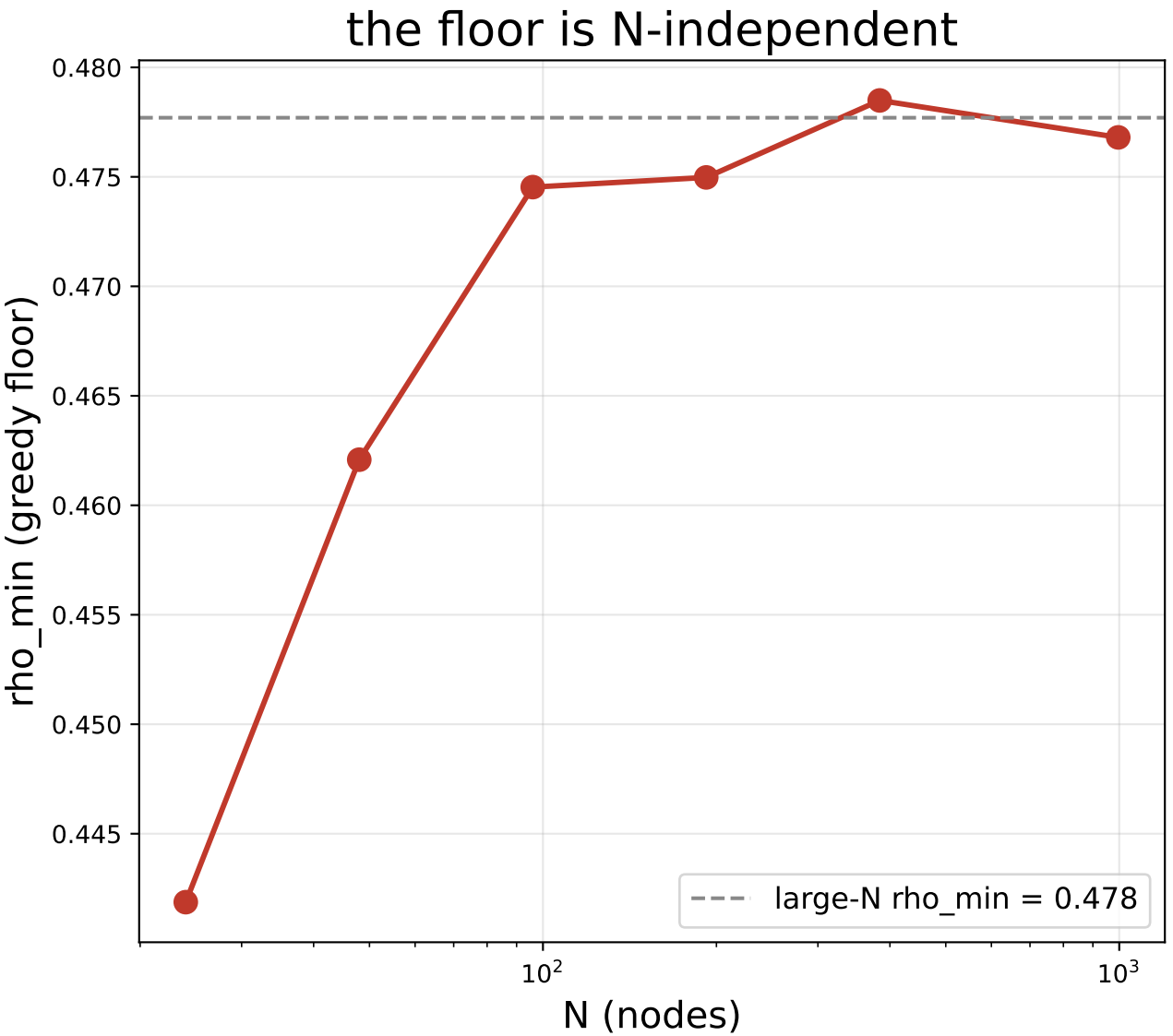
DD-Sudoku transformer

hint-density transition + what algorithm it uses

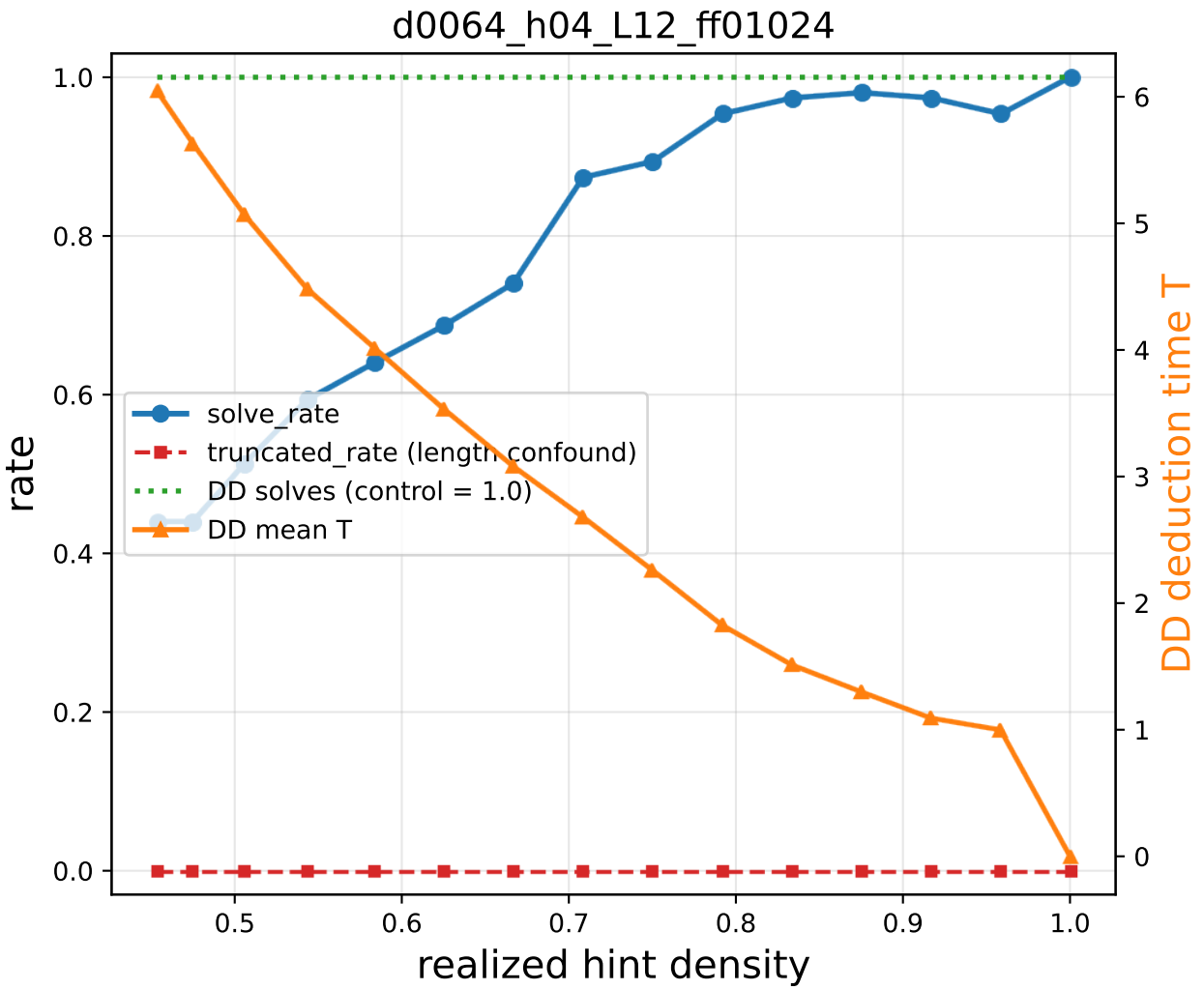
The transition IS there: solve_rate 0.99 -> 0.44 as DD's deduction depth goes 1 -> 6.
But the model is not running DD, and never starts: bookkeeping is in place by epoch 5, before it can solve anything,
while the deducibility predicate stays flat for 895 more epochs. What training buys is which COLOR to place, not which node is deducible.

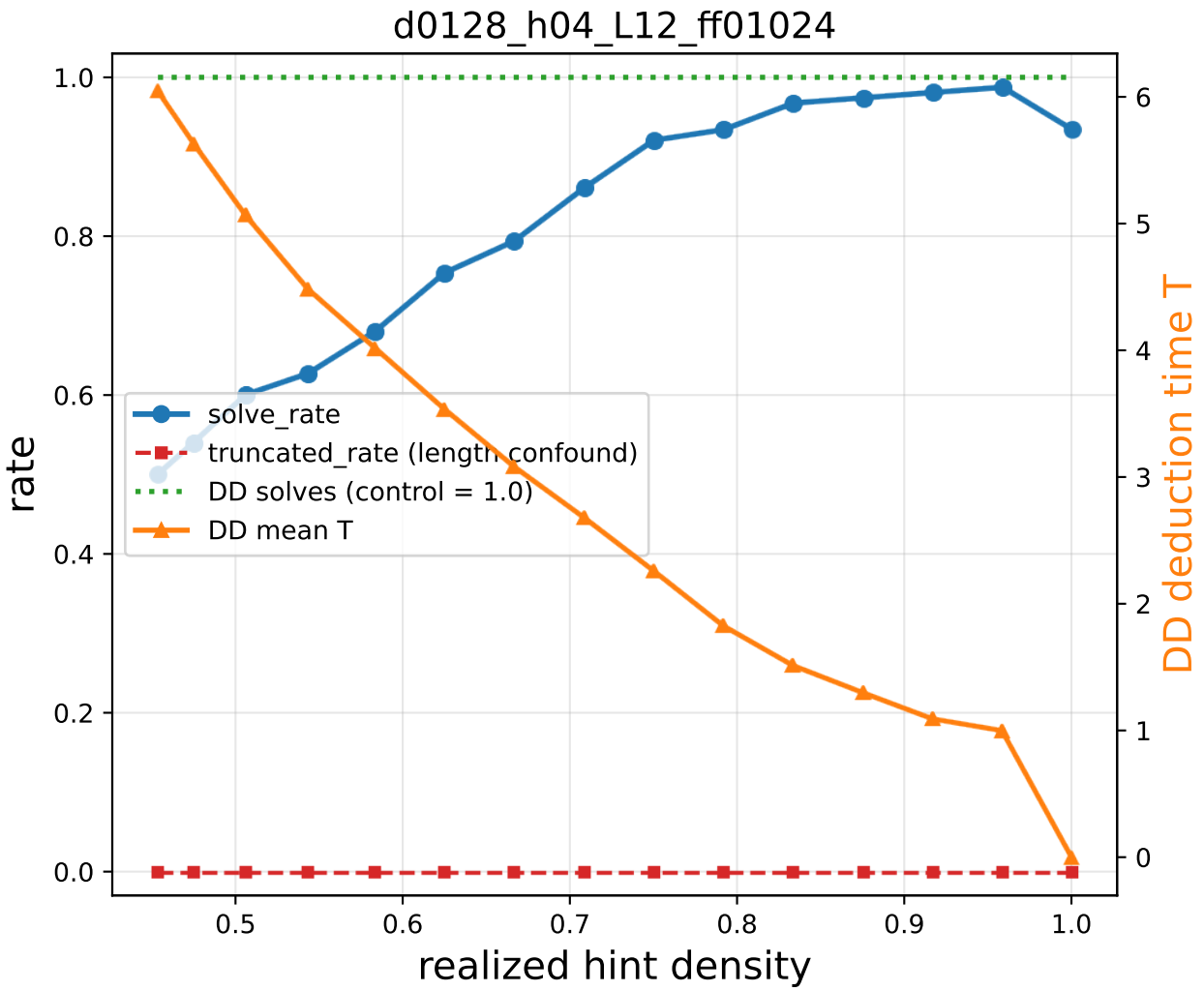
Where the transition is: DD deduction time diverges at rho_min





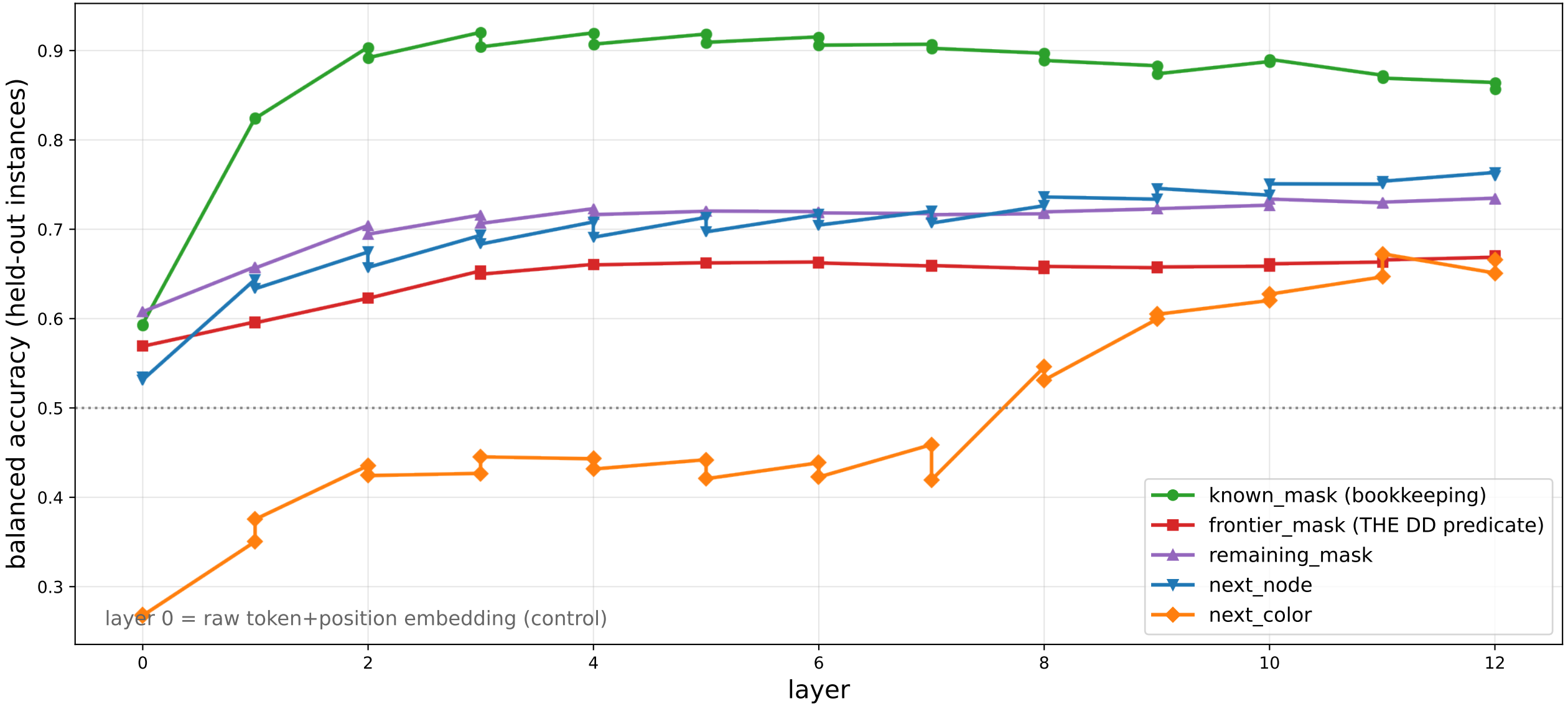
Does the model collapse where DD's reasoning depth diverges?



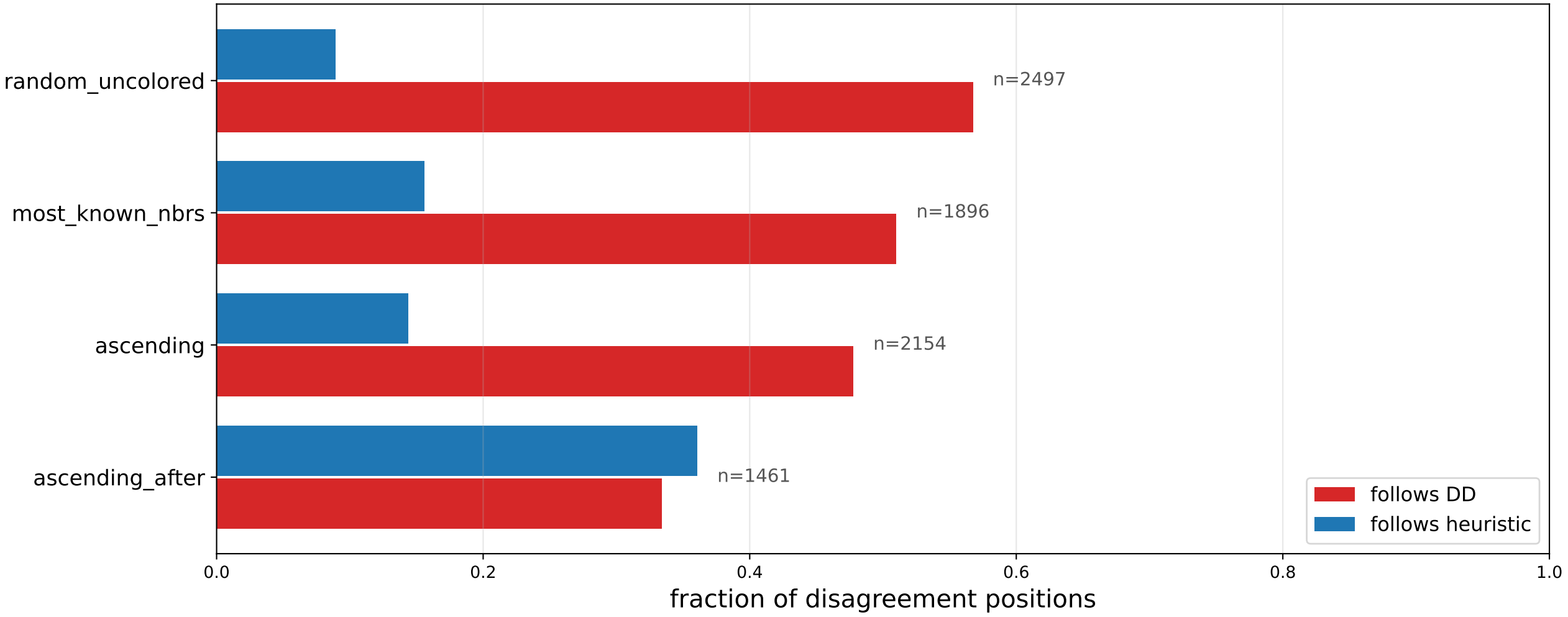


CAVEAT: rho and T are collinear in this ensemble (removing hints is what lengthens the chain), so r(solve,T) = -0.97 vs r(solve,rho) = +0.94 cannot attribute the collapse to depth rather than density.

Probes: bookkeeping yes, DD predicate no

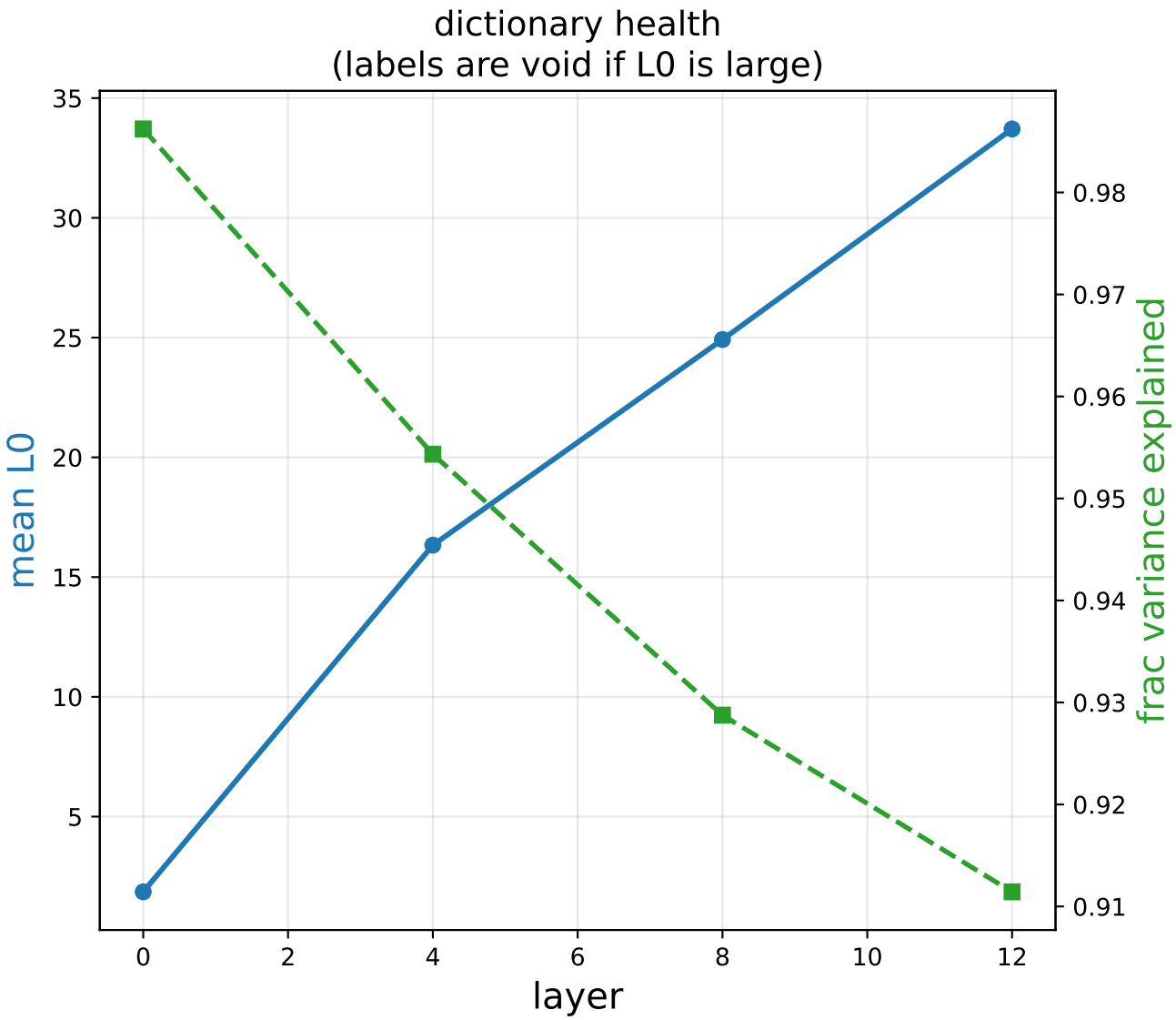


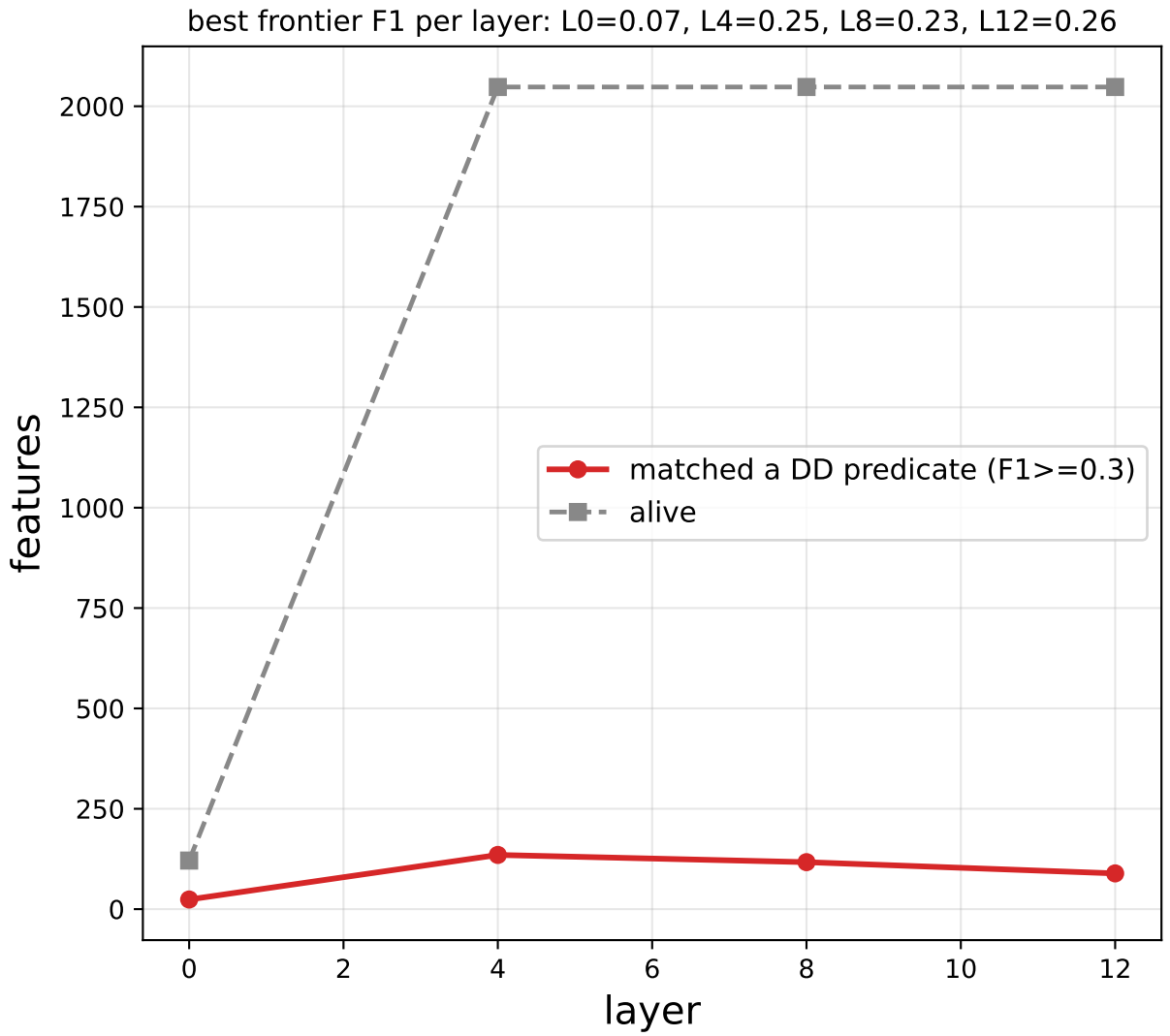
Which rule wins where DD and a heuristic disagree?



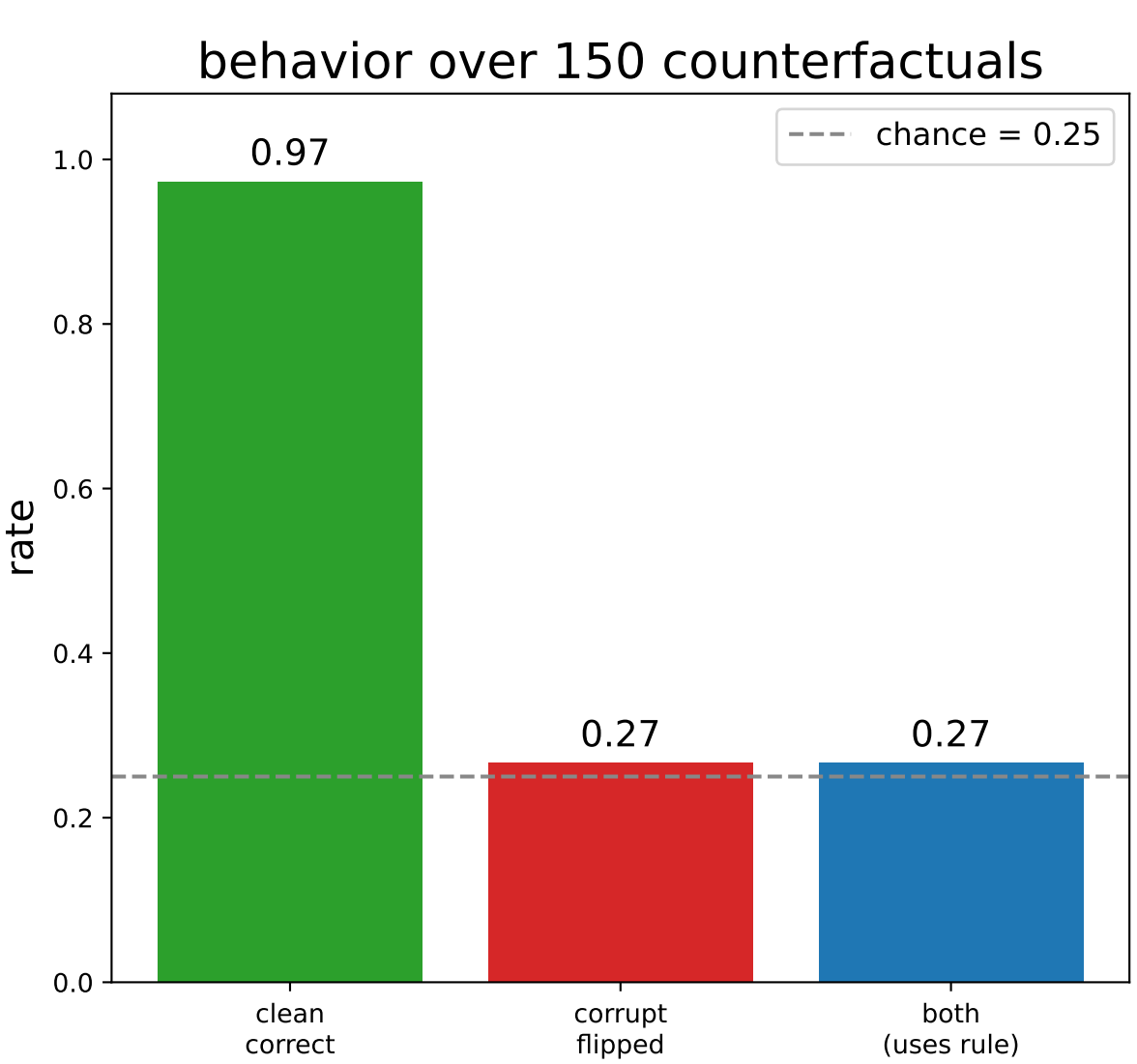
ascending_after = 'next uncolored id above the last emitted, wrapping'
the only baseline that exploits the sorted-emission convention -- and it beats DD

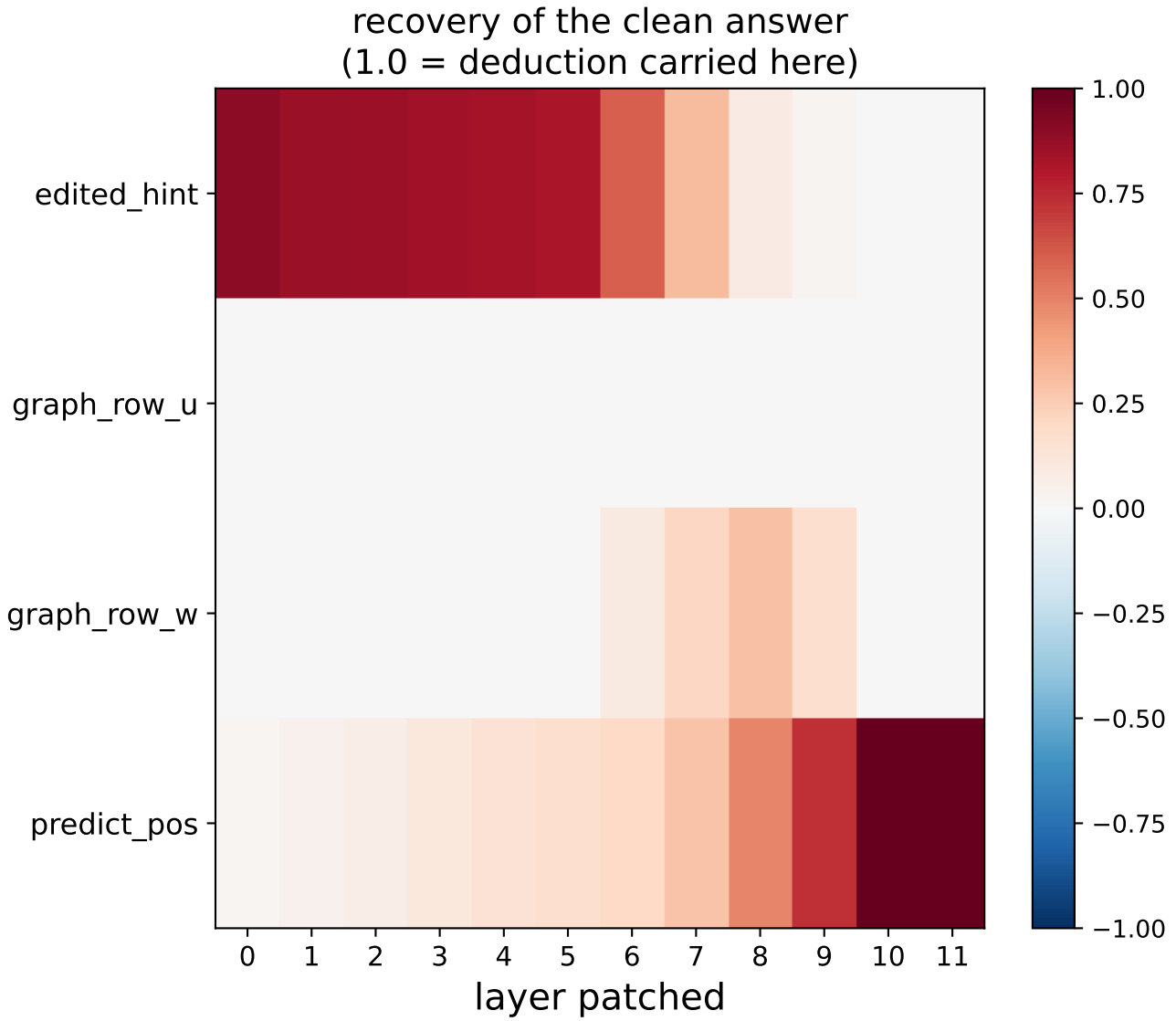
SAE: do sparse directions individually mean DD predicates?



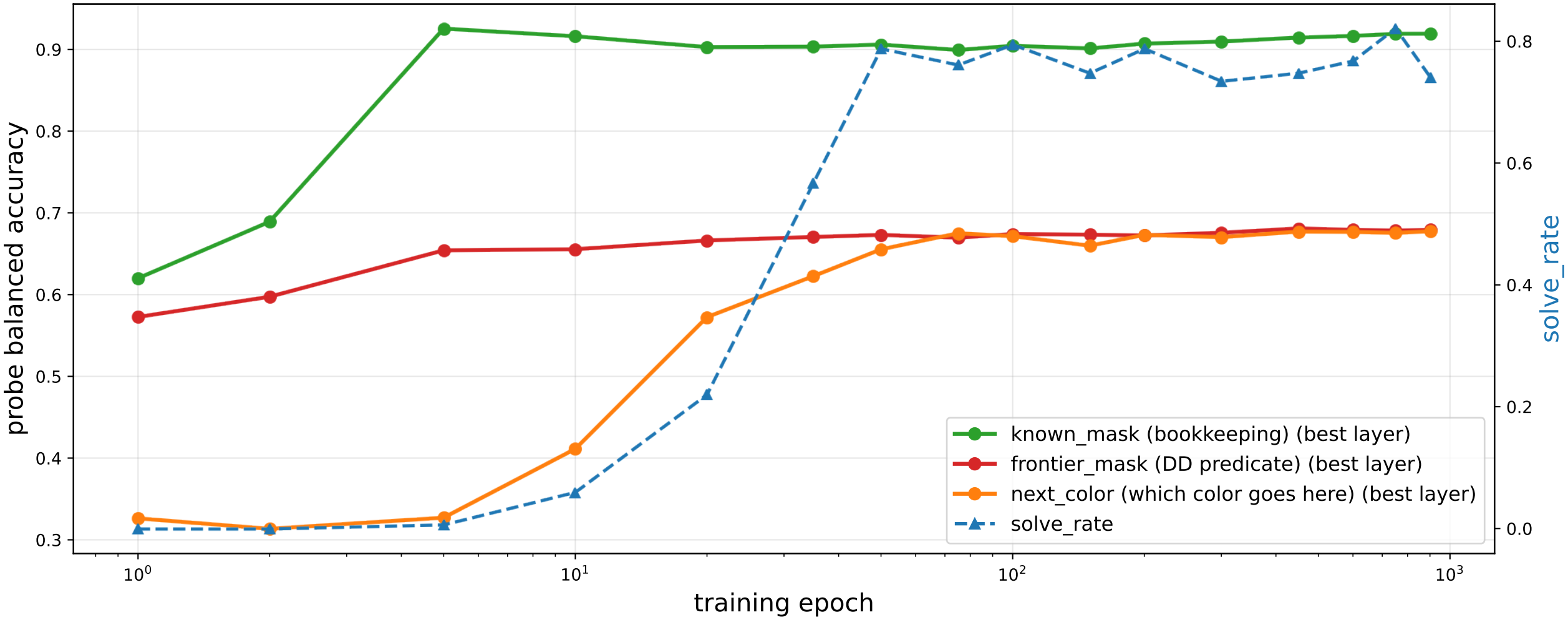


Causal test: one neighbour-color token flips DD's answer





When during training does DD structure appear?



Answer: never. Bookkeeping is done by epoch 5, before the model can solve anything (solve_rate 0.007); the DD predicate then moves 0.654 -> 0.679 over 895 more epochs while solve_rate grows 100x. What develops with competence is next_color (0.327 -> 0.655): the node is picked by an ordering convention learned in 5 epochs, and training goes into choosing the colour.

Summary and what it means

1. The hint-density transition is NOT in independent random hints -- that ensemble is subcritical at every density, with recovery identical from N=24 to N=384. It is in the removal-path ensemble the models already train on, as a divergence of DD's deduction time T as rho approaches the floor rho_min = 0.478 (N-independent).

2. The gradient exists at N=24, so the transition is testable on existing checkpoints. This retires the earlier plan to scale N to ~1700 tokens. MEASURED: solve_rate falls 0.99 -> 0.44 monotonically as DD's depth rises T=1.0 -> 6.0, with truncated_rate = 0.000 in every bin, so it is not a generation-budget artifact. But rho and T are collinear here, so which of the two controls the collapse is NOT yet resolved -- that needs equal-rho instances binned by T.

3. Probes: bookkeeping (known_mask, 0.92) is computed early and strongly; DD's deducibility predicate (frontier_mask, 0.67 vs a 0.57 control) is barely present. next_color is computed late (0.46 at L7 -> 0.65 at L12). CAVEAT: the frontier_mask number should not be read as a weak DD trace at all -- the deducibility-free rule 'not known' scores 0.73 on the same metric, above the model, while a procedure-matched probe on the known bits scores 0.59, below it. Balanced accuracy on a 15%-positive target is miscalibrated both ways; needs an AUC rescore.

4. Causal test: the model does NOT apply the forcing rule at decision level (flip rate 0.27 vs 0.25 chance), but neighbour-color information does flow: the corruption moves the logit gap by 12.3. Patching localizes that flow to edited_hint (L0-5) -> graph row of the RECOLORED NEIGHBOUR (L6-9) -> prediction position (L8-11). The row of the node being predicted carries ~0.00 throughout, so this is color-to-node binding, not DD's neighbour-list join.

5. When during training: NEVER. Bookkeeping reaches 0.93 by epoch 5, before the model can solve anything (solve_rate 0.007); the DD predicate then moves 0.654 -> 0.679 over 895 more epochs while solve_rate grows 100x. The one target that develops WITH competence is next_color (0.33 -> 0.66, r=0.96 with solve_rate), while next_node is finished by epoch 5 like the rest of the bookkeeping. Best current description of the algorithm: pick the next node by